

MD. PULAK BHUIYAN

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Academic Credential

Shahjalal University of Science and Technology (SUST), Bangladesh

Bachelor of Science (B.Sc.)

Mechanical Engineering

2020 – 2025

CGPA: 3.70/4.00

Major Courses: Engineering materials, Solid Mechanics, Thermodynamics, Fluid Mechanics, Fluid Machinery, Heat Transfer, Machine Design, Mechanics of Machinery, IC Engine, Automobile Engineering, Basic Mechatronics, Powerplant Engineering.

Research Interests

Heat and Mass Transfer

Thermal Engineering

CFD

Fluid Mechanics

Machine Learning

Research Experience

Undergraduate Thesis

(2025)

Topic: Numerical Investigation on Enhancing the Thermal Performance of Micro-Tesla Valve Using Pin-fins.

Supervisor: HM Toufik Ahmed Zisan, Department of MEE, SUST

- A numerical study on microchannel heat sink with a two-stage micro-Tesla valve where micro pin-fins have been inserted in them to maximize the heat transfer efficiency and to optimize the diodicity of the valve by using SST $k-\omega$ turbulence model. The pressure drop penalty for adding fins have also been considered for optimizing the overall performance of the design.

Research Assistant | SUST Research Center

(Feb 2024 – Mar 2025)

Project: Development of an Autonomous Robot for Floating Solid Waste Detection and Collection from Rivers Using Machine Learning Algorithms.

Supervisor: Md. Mahmud-Or-Rashid and Md. Syamul Bashar, Department of MEE, SUST.

- Collaborated with a team for on-going funded projects under Mechanical Engineering department.
- Performed in-depth literature review, analyzed several object detection algorithms and identified key research gaps.
- Assemble, test and improve different project prototypes through extensive field testing.

Conference Paper

Paper Title: A Comparative Study on Enhancing the Thermal Performance of a Micro-Tesla Valve Using Pin-Fins.

Authors: Md. Pulak Bhuiyan, Muhaammad, HM Toufik Ahmed Zisan

Conference: 3rd International Conference on Mechanical, Manufacturing and Process Engineering (ICMMPE 2026)

Projects

Chem-E-Car

(Dec 2024)

- A self-constructed car which can operate with the electric power generated from a chemical reaction and have a stopping mechanism depending on another chemical reaction after going a certain distance.

Advance Line Follower Robot

(Feb 2023)

- An LFR made with hand-made PCB which can follow a black line on white surface and vice versa. It's able to make decisions in critical situations like avoiding obstacles, solving maze, going inside of a cave, passing bridges and seesaws etc.

3rd Year Project: SonarSLAM

(2023)

- A four wheeled autonomous robot which can create a 2D map of an unknown environment using ultrasonic sonar sensors made with Raspberry Pi. By using ROS2 and Extended Kalman Filter algorithm, SLAM (Simultaneous Localization And Mapping) has been implemented in the bot.

Bumperbot

(2024)

- A simulated robot on Gazebo environment using ROS2 that can do localization and mapping of a closed environment with SLAM Toolbox also can be controlled using a joystick connected with the computer externally.

Internship and Training

Industrial Training

(Jan 2024)

Bangladesh Steel Re-Rolling Mills Ltd. (BSRM), Nasirabad, Chittagong

- Acquired 15-days hands-on training experience in rolling mill operations and equipment maintenance.
- Learned about quality control processes and safety protocols to enhance operational efficiency on steel production.

Industrial Tour

(21 Aug, 2023)

Chevron Jalalabad Gas Field, Lakkatura, Sylhet

Industrial Visit

(11 Mar, 2023)

Sylhet 225MW Combined Cycled Power Plan (CCPP)

Leadership and Volunteer Experience

General Secretary

(Apr 2024- Jun 2025)

RoboSUST, Sylhet

- Lead the voluntary robotics organization and organized five workshops and two competitions.
- Conducted “Mythical RoboSUST Month 2024”, a month-long robotics event for 100 new robotics enthusiasts.

Assistant Team Leader

(Sep 2018- Jun 2019)

Red Crescent Youth, Cumilla

Technical Skills

- **3D Design** : SolidWorks (CSWA, CSWA-S, CSWA-3D Mold Creator)
- **Simulation** : COMSOL Multiphysics, Ansys, Simulink
- **Programming** : Python, C, C++, Reinforcement Learning, MATLAB, C-make, XML, Git & GitHub
- **Operating System** : Windows, Ubuntu Linux
- **Robotics & Hardware** : ROS 2, Gazebo, Arduino, Raspberry Pi, TinkerCAD, Circuit Design, EasyEDA
- **Others** : MS Office, Canva, Supply Chain Management, Tecplot 360, Data Manipulation

Soft Skills

Leadership	Teamwork	Problem Solving	Time Management	Flexibility
Event Management	Communication	Adaptability	Public Speaking	Critical thinking

Awards and Achievements

- **2nd Runner-Up** - CHEM-E-CAR Competition, BUET (06 Dec 2024)
- **Runner-Up** - Spot-N-Go (LFR competition), Init 3.0 (04 Mar 2023)
- **Regional Winner** - 8th Bangladesh Physics Olympiad 2018 (Dec 2018)
- **2nd on General grade** – Future Leader Scholarship, Cumilla Model College, Cumilla (25 Nov 2016)